THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

This exam paper must not be removed from the venue

Venue _____
Seat Number _____
Student Number |__|__|__|__|__|__|__|__|
Family Name _____
First Name _____

School of Mathematics & Physics
Semester Two Examinations, 2024
PHYS7045 Quantum Technologies

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

•Any written or printed material is permitted; material may also be annotated.
•Casio FX82 series or UQ approved and labelled calculator only
•During Planning Time - Students are encouraged to review and plan responses to the exam questions

Materials Permitted in the Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)

Any additional written or printed material is permitted; material may also be annotated.

Materials to be supplied to Students:
Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

1 x 14-Page Answer Booklet

Instructions to Students:
If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

For Examiner Use Only

Question	Mark
1	
2	
3	

Total _____

Question 1.

A code defined on five qubits has the following stabiliser generators

$$S_1 = Z\ Z\ Z\ I\ I$$
$$S_2 = I\ I\ Z\ Z\ Z$$
$$S_3 = X\ I\ X\ X\ I$$
$$S_4 = I\ X\ X\ I\ X$$

Part A (grade: 1-5)

1. Do you expect S_1 to commute with all the other stabilisers listed in the table, or not? Explain your answer. Confirm whether this is true (and show your reasoning).
2. A logical X_L operator is defined for this code as $X_L = X\ X\ I\ I\ I$. Verify whether this commutes with the stabilisers.
3. Is this code able to detect a single qubit bit-flip error (i.e. an X error) on any one qubit? Give reasons based on possible syndromes. Can it correct such an error?
4. Is this code able to detect a single qubit phase error (i.e. a Z error) on any one qubit. Give reasons based on possible syndromes. Can it correct such an error?
5. Explain whether the code can correct any single qubit X, Y or Z error? If so, what is the lower bound on the code distance?

Part B (Advanced: grade +0,+1,+2)

a) Find a logical Z_L operator for the code, and verify whether it has the correct commutation and anticommutation relations with the stabilisers and with X_L.

b) Calculate the code distance, and explain whether this is consistent with questions A3, A4 and A5 above.

Question 2.

A bosonic qubit, in dual rail format, is encoded in the tensor product Hilbert space of two modes as an eigenstate of total photon number with eigenvalue 1:

$$|\mathbf{0}\rangle = |1\rangle \otimes |0\rangle$$

$$|\mathbf{1}\rangle = |0\rangle \otimes |1\rangle,$$

where $|n\rangle$ is a photon number eigenstate.

Part A (grade: 1-5)

1. Given the unitary 'beam-splitter' transformation for two modes:

$$a \to a\cos\theta + be^{i\phi}\sin\theta$$

$$b \to b\cos\theta - ae^{-i\phi}\sin\theta$$

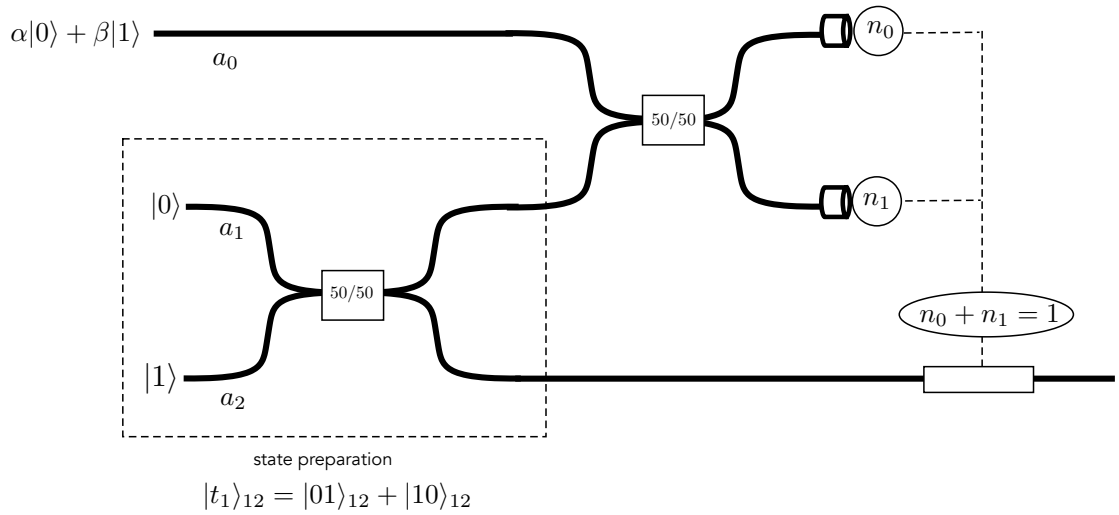
$$a^\dagger \to a^\dagger\cos\theta + b^\dagger e^{i\phi}\sin\theta$$

$$b^\dagger \to b^\dagger\cos\theta - a^\dagger e^{-i\phi}\sin\theta,$$

where $a, b, a^\dagger, b^\dagger$, are the annihilation and creation operators for the two modes. Show how to implement a Hadamard gate, in the logical basis, with a suitable choice of θ, ϕ.

Part B (Advanced: grade +0,+1,+2)

It is not possible to do teleportation with dual rail encoded qubits using linear optics alone. However, a non-deterministic scheme exists and can be constructed for single photon states.

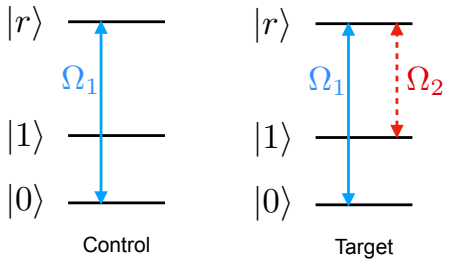


a) Prove that teleportation of the input state $\alpha|0\rangle + \beta|1\rangle$ works 50% of the time, provided the total count at the top two detectors equals one.

Question 3.

The two atom Rydberg CNOT gate described in lectures used a series of sequential applications of localised pulses to the two atoms. Since there are technical challenges in implementing this approach, the use of simultaneous pulses can provide technical advantages.

In the approach shown in the figure, a simultaneous $|0\rangle \rightarrow |r\rangle$ pulse Ω_1 is applied to two adjacent atoms, and a second $|r\rangle \rightarrow |1\rangle$ pulse Ω_2 is simultaneously applied to the second atom only. Note that applying the two wavelength pulses simultaneously results in a two-photon transition, coupling the $|1\rangle$ and $|0\rangle$ states.



Part A (grade: 1-5)

1. What are the pulse areas needed ($2\pi, \pi$, etc.) needed for the control and target pulses? Explain your answer.
2. Assuming double-excitation to the Rydberg state is blockaded, show the effect of applying the pulse sequence on the qubit basis states $|1_c 0_t\rangle, |1_c 1_t\rangle, |0_c 1_t\rangle, |0_c 0_t\rangle$, demonstrating that this results in a CNOT gate. Explain your reasoning.
3. In comparison to the scheme described in class, describe one technical advantage of this approach to realising a two-qubit gate with Rydberg atoms.

Part B (Advanced: grade +0,+1,+2)

a) Show how the above scheme can be used to realise a Toffoli gate with three qubits and the same global pulses, with Ω_2 again applied to the target only. *Hint: consider the geometric arrangement of the atoms, noting that the Rydberg blockade applies to nearest neighbours.*

END OF EXAMINATION